Hadoop & Spark - ISEN 2025/2026

Lab: Pair RDDs

For this Lab you will need:

- The 3.2 Lab

Pair RDDs

You can create a pair RDD by creating a RDD with (key, value) pairs

```
import math

numbersRDD = sc.parallelize([1,2,3,4,5,6])

# Create Pair RDD
squareRootRdd = (
    numbersRDD.map(lambda x: (
        x,              # <- Key
        math.sqrt(x)    # <- Value
    )
    )
)

squareRootRdd.collect()
```

Reminder

Get this snippet to reload the data if necessary for exercise 1 through 3

```
rddFile = sc.textFile('lego_sets_and_themes_clean.csv')
header = rddFile.first()
rddFile = rddFile.filter(lambda row: row != header)
rddFileWithCols = (
    rddFile.map(lambda line: line.split(";"))
)
rddFileWithCols.take(3)
```

Exercise 1

Try to find the number of sets for each theme in our LEGO RDD (theme is the 5th column), by using pair RDDs.

Please name your RDD `themeCountRDD` and check the following slide for help.

> **Hints** First create an RDD with a map (key,value) where keys are the themes of each line, and the value is 1. Then use the `reduceByKey(lambda: ...)` method to apply a reduce.

reduceByKey documentation

Common useful functions for Pair RDDs (1/2)

Sort by keys:

```
themeCountRDD.sortByKey().take(5)
```

List keys (returns duplicates as well)

```
themeCountRDD.keys().take(5)
```

Common useful functions for Pair RDDs (2/2)

List values (returns duplicates as well)

```
themeCountRDD.values().take(5)
```

Distinct returns the unique (key,value) pairs

```
themeCountRDD.distinct().take(5)
```

Exercise 2

Find the average number of parts per set for each LEGO theme using pair RDDs.

You will need to filter the invalid numbers of parts first.

Hints:

- 1 First, create another RDD with a map (key, value) where keys are the themes of each line, and the value is always 1 (like previous exercise).
- 2 Then, create an RDD with a map (key, value) where keys are the themes of each line, and the values are the number of parts in each set. (For example {'Technic', 3} for the first element with Technic the set and 3 the number of elements)
- 3 Use the reduceByKey(lambda: ...) method on both RDDs to create two separate RDDs with the sum of parts and the number of sets per theme.
  Note: For this step, if you have a value error on 'int', try to make a filter on rows with invalid number of parts (such as empty or not int)
- 4 Use the join() method to join both RDDs on the theme keys.
- 5 Calculate the average number of parts per set for each theme.